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S&P Global & CRISIL Campus Hackathon 2026

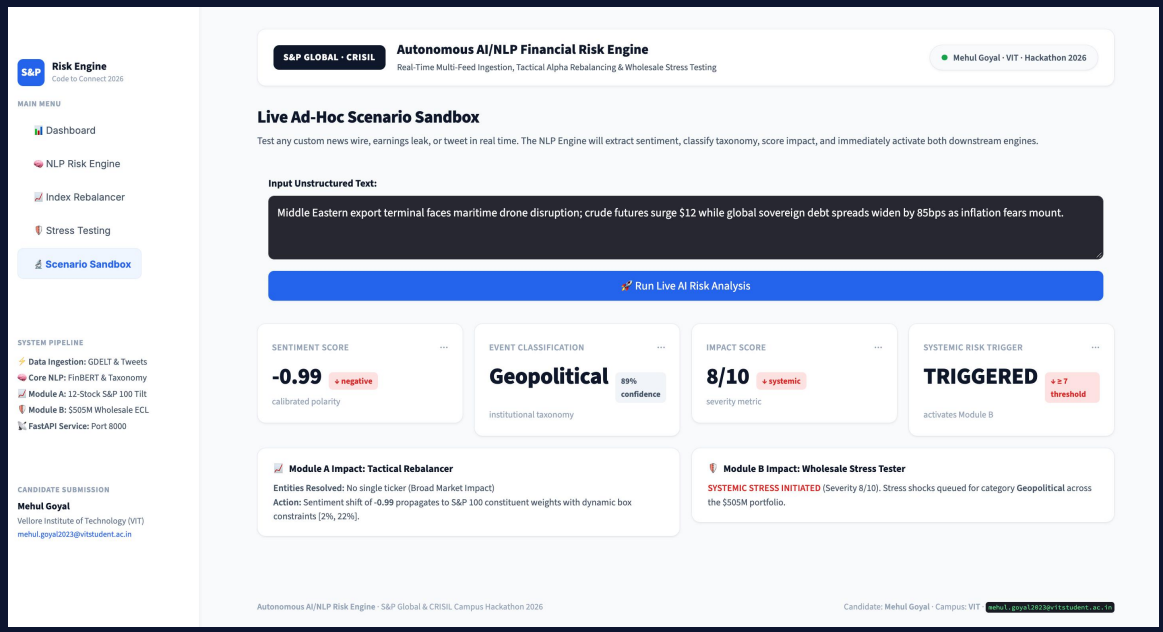
AUTONOMOUS AI/NLP FINANCIAL RISK ENGINE

Autonomous AI/NLP Financial Risk & Intelligence Platform

Unstructured news & social text in. Sentiment, event class and impact score out, driving a portfolio rebalancer and a bank-book stress test.

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LIVE SCENARIO SANDBOX

Real-Time Multi-Feed Ingestion · Tactical Alpha Rebalancing · Wholesale Stress Testing

EVENT TAXONOMIES
5
Geo · Macro · Credit · M&A · Product

SIGNALS INGESTED
12
6 news wires + 6 social posts

DOWNSTREAM ENGINES
2
Module A (equity) + Module B (credit)

WHOLESALE BOOK
14
facilities · $550M notional

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Problem Statement & FinTech Innovation Framework

Manual reading cannot keep pace with news-driven shocks, so we turn text into decisions automatically

THE PROBLEM

01 Unstructured overload

News wires, filings and social feeds arrive faster than analysts can read them.

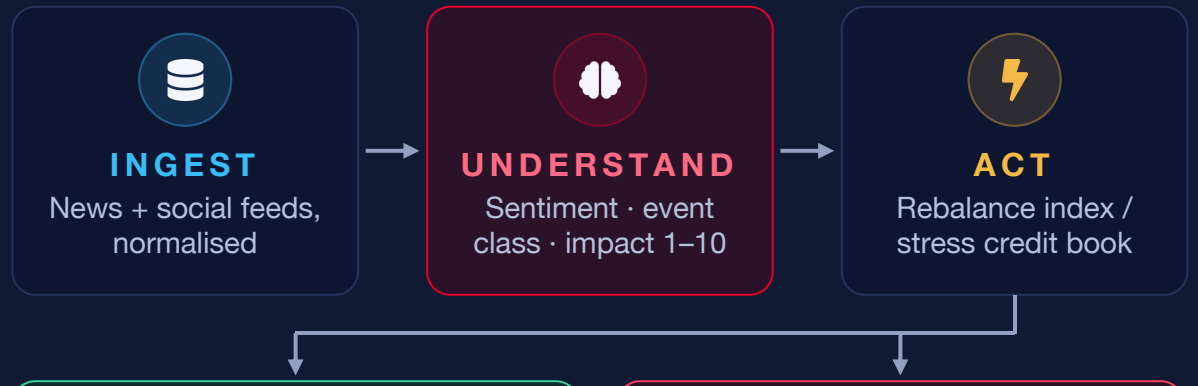
02 Missed alpha

Sentiment moves prices within minutes; static index weights ignore it.

03 Late credit warning

Quarterly reviews are backward-looking, so systemic shocks hit the loan book unseen.

OUR FRAMEWORK: ONE SIGNAL, TWO DECISIONS



Innovation: a single NLP signal serves both asset management and wholesale credit risk, so the same news that moves an index also stresses a loan book.

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Multi-Feed Ingestion & NLP Risk Classification Engine

Two text feeds are normalised and scored into three machine-readable risk signals

News wire feed
6 items · GDELT / Reuters-style

Social sentiment feed
6 posts · cashtags ($AAPL, $JPM)

Ingestion

12

Unified schema: id, channel, timestamp, text, entities

AI/NLP RISK ENGINE

Sentiment
–1 to +1
Weighted financial lexicon, negation lookback, tanh scaling

Event class
5 taxonomies
Keyword density + softmax gives a confidence

Impact
1 to 10
Category severity + polarity + magnitude + breadth

WORKED EXAMPLE FROM THE LIVE SANDBOX

> *Middle Eastern export terminal faces maritime drone disruption; crude futures surge $12 while global sovereign debt spreads widen by 85bps as inflation fears mount.*

SENTIMENT SCORE
-0.99 negative
–1 0 +1

EVENT CLASSIFICATION
Geopolitical 89%
89% confidence

IMPACT SCORE
8/10 systemic

SYSTEMIC RISK TRIGGER
TRIGGERED
≥ 7 threshold

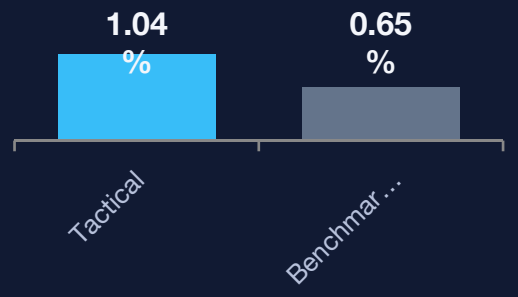
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Tactical Portfolio Rebalancing (Module A) & Constrained Optimization

Sentiment tilts a 12-stock S&P 100 basket inside institutional box constraints

TACTICAL VS BENCHMARK RETURN

+38.8 bps alpha



TACTICAL SHARPE

0.24

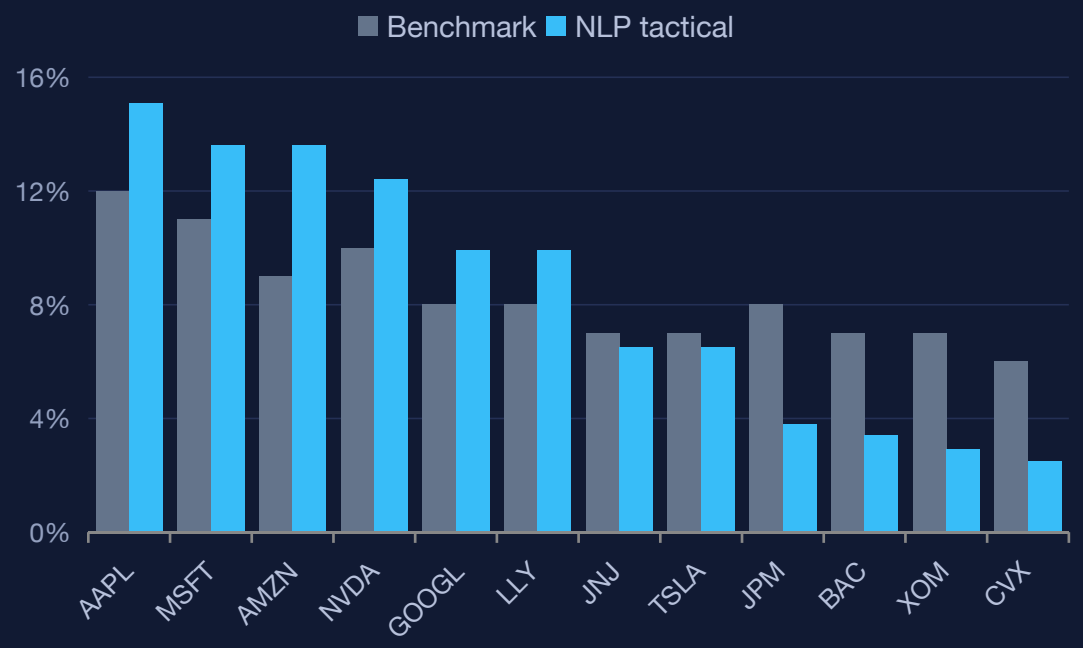
Risk-adjusted

ANNUALISED VOLATILITY

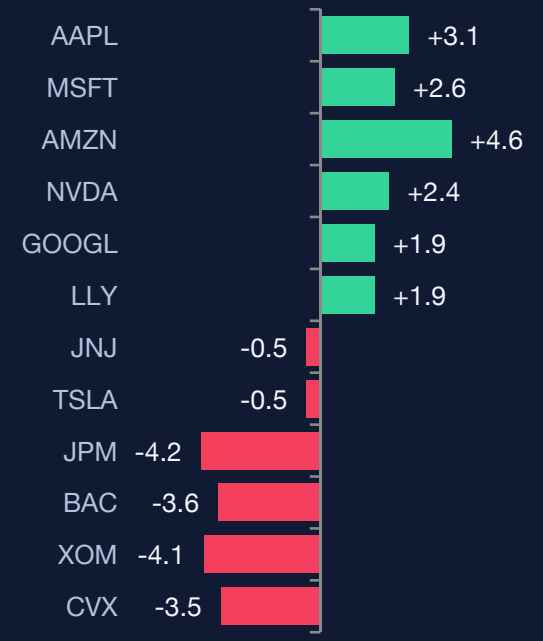
6.67%

Controlled risk

WEIGHT ALLOCATION (%): BENCHMARK VS NLP-TILTED



SHIFT VS BENCHMARK (PP)



HOW IT WORKS

1. **Aggregate** — net sentiment per ticker, weighted by impact
2. **Tilt** — $w' = w \times (1 + \lambda \cdot s)$, with $\lambda = 0.65$
3. **Constrain** — 2% floor, 22% cap, sum = 100%

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Wholesale Portfolio Stress Testing (Module B – Basel EL Framework)

Impact ≥ 7 triggers category-specific shocks across loans, bonds and derivatives

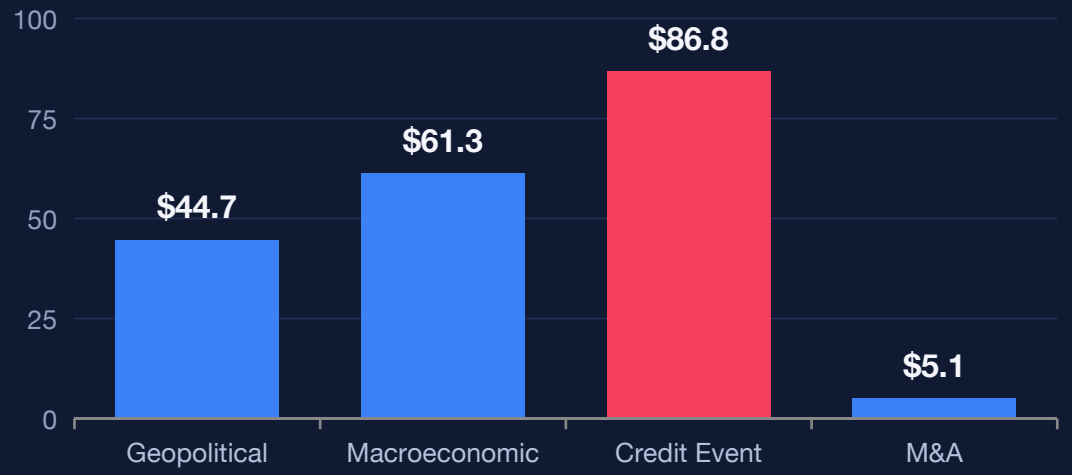
PRE-STRESS VALUE
$545.5M
14 facilities

POST-STRESS VALUE
$505.4M
↓ -7.35%

EXPECTED LOSS
$11.71M
↑ +161% surge

SHOCK MULTIPLIER
0.67x
Macroeconomic

PORTFOLIO VALUATION LOSS BY SCENARIO ($M, IMPACT 8)



SHOCK MATRIX AT FULL SEVERITY

Event category	Spread +bps	PD ×	LGD +
Geopolitical	160	1.75	15%
Macroeconomic	120	1.90	10%
Credit Event	280	3.10	35%
Merger / Acquisition	40	1.15	5%
Product Launch	20	1.05	2%

EL = EAD × PD × LGD · shock matrix at full severity

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Technology Architecture, Verification Results & Business Impact

A reproducible, offline prototype that runs end to end with one command

ARCHITECTURE & STACK

1 **News + social feeds**
JSON · GDELT/StockTwits-style

→

2 **Ingestion & NLP engine**
Python 3.11 · regex · lexicon

→

3 **Module A | Module B**
NumP y · Pandas · SciPy

→

4 **Streamlit UI + FastAPI**
Plotl y · Uvicorn · Pydantic

VERIFICATION RESULTS

~0.1 ms
per document through the NLP engine



7 of 12 signals flagged systemic (impact ≥ 7)

100%



weights sum to 1; 2.5% to 15.1%, inside the 2–22% box

Deterministic
seeded data; reruns give identical results

BUSINESS IMPACT

Early-warning credit intelligence
continuous surveillance instead of quarterly reviews

Smart-beta sentiment indices
weights that adapt to market regimes

Regulatory stress readiness
instant shock transmission, Basel / CCAR style

Limitations & next steps: synthetic data and prices; lexicon NLP today, FinBERT upgrade next; shock parameters are assumptions; plug in live feeds and calibrated PD/LGD.